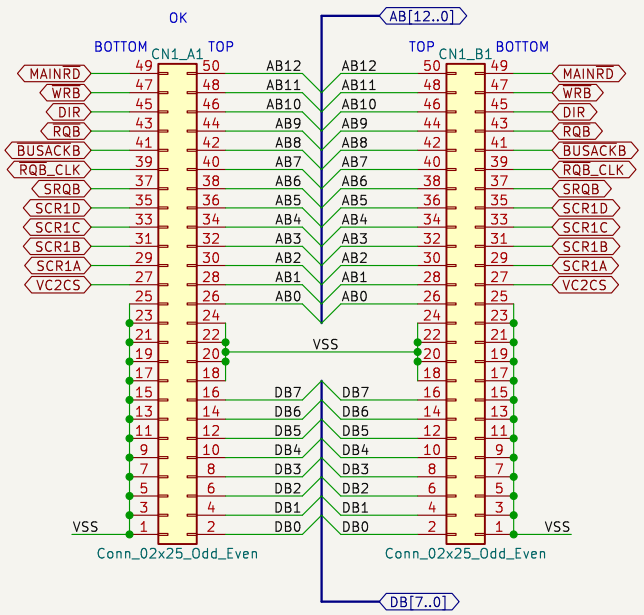
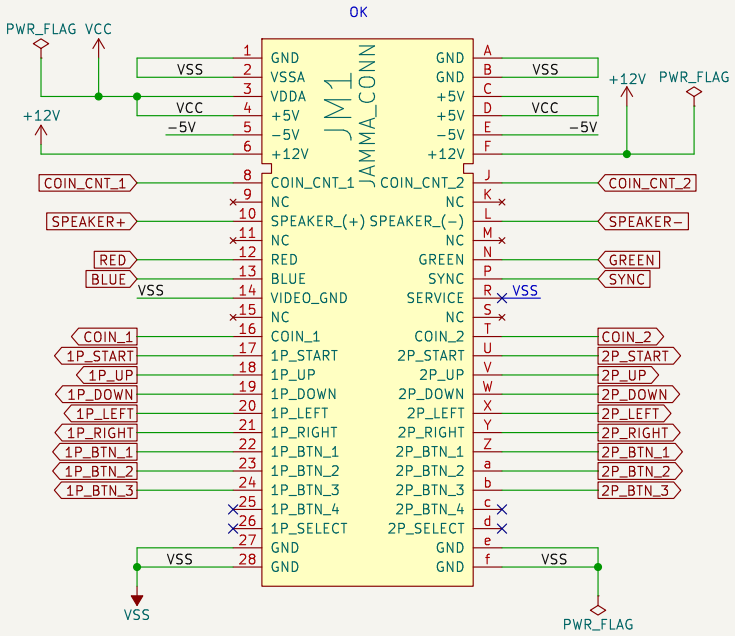
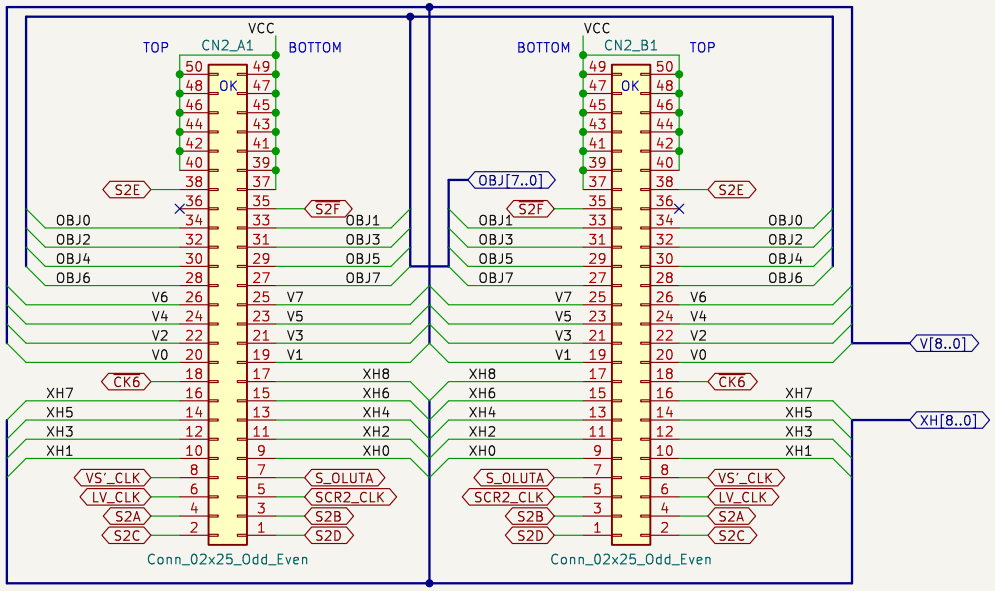






Observation
The pcb that has the jamma connector is the upper level and the location of each device is labeled with number, letter and number1. On the lower level pcb, the devices are labeled with letter and number.

video
File: video.kicad_sch

sound
File: sound.kicad_sch

main
File: main.kicad_sch

interruption
File: interruption.kicad_sch

video_timings
File: video_timings.kicad_sch

v_counter
File: v_counter.kicad_sch

h_counter
File: h_counter.kicad_sch

last_u
File: last_u.kicad_sch

capacitors
File: capacitors.kicad_sch

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JOTEGO

Sheet: /
File: exed.kicad_sch

Title: *Exed Exes board B*

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 1/28

v_scroll

File: v_scroll.kicad_sch

char

File: char.kicad_sch

h_scroll

File: h_scroll1.kicad_sch

object_table

File: object_table.kicad_sch

scroll1

File: scroll1.kicad_sch

obj_ram

File: obj_ram.kicad_sch

scroll2

File: scroll2.kicad_sch

obj_lut

File: obj_lut.kicad_sch

colmix

File: colmix.kicad_sch

obj_counters

File: obj_counters.kicad_sch

logics_gates

File: logics_gates.kicad_sch

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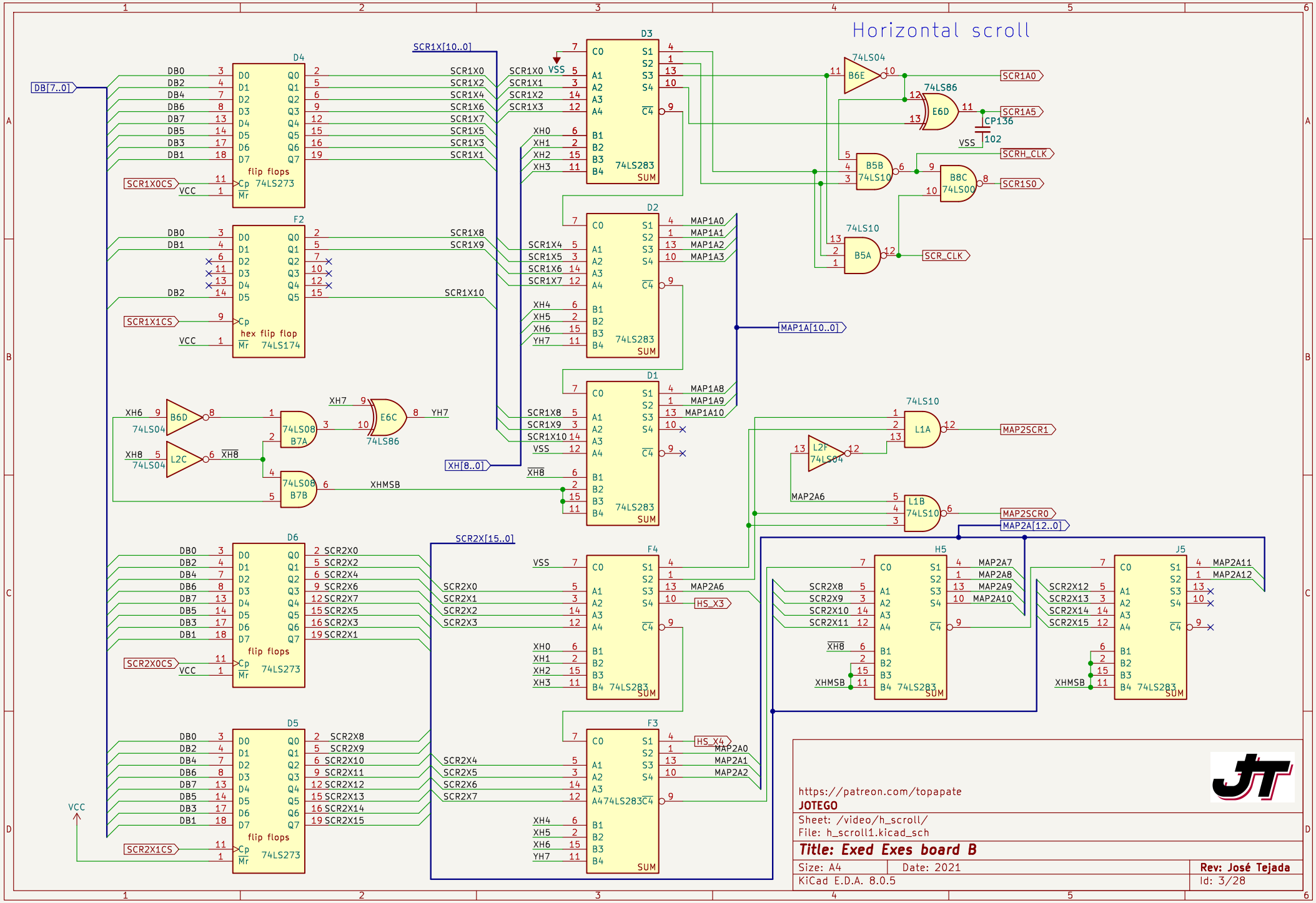
JOTEGO

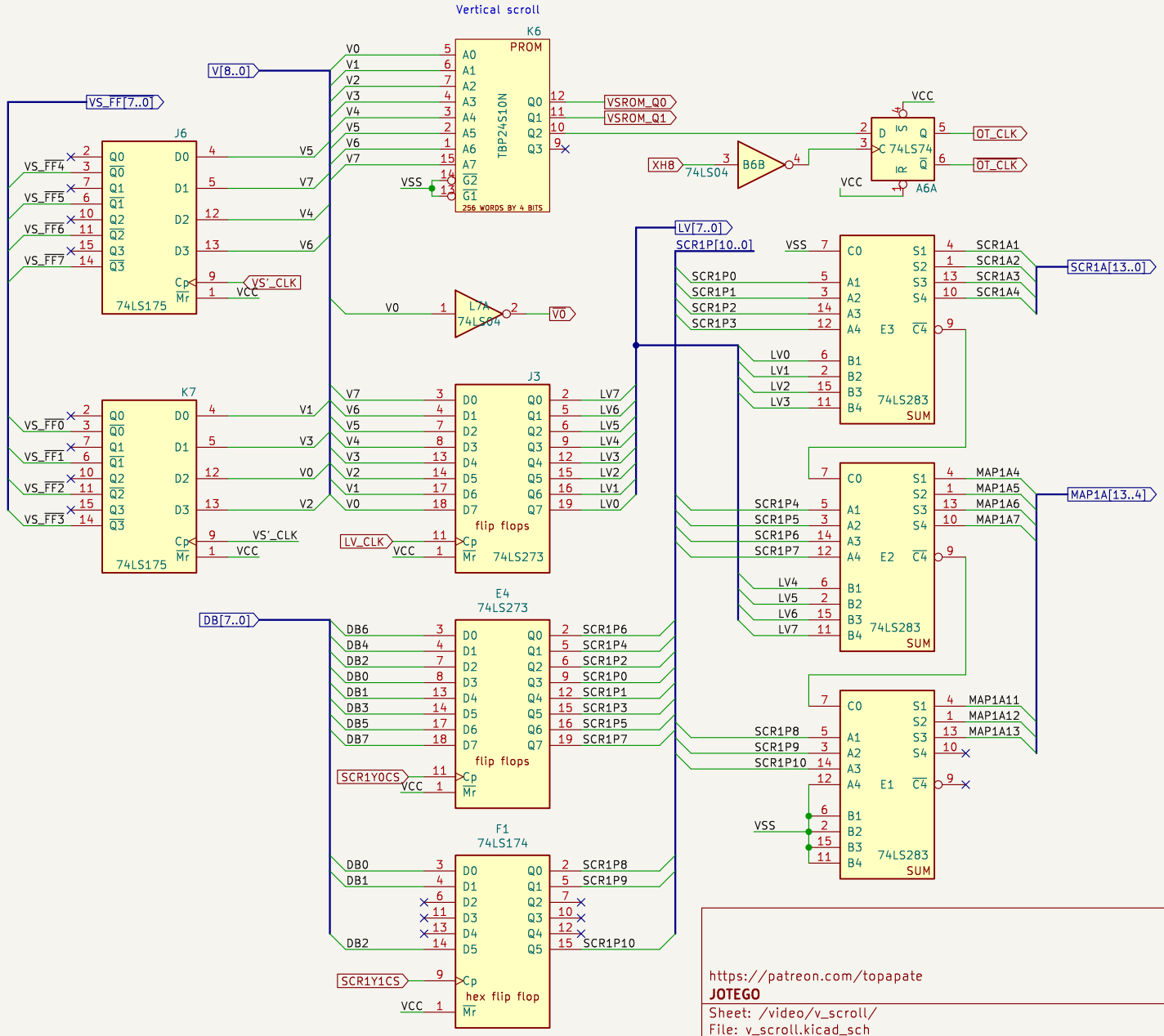
Sheet: /video/

File: video.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 2/28


Horizontal scroll
DB[7..0]
SCR1X[10..0]
SCR2X[15..0]
MAP1A[10..0]
MAP2A[12..0]
XH[8..0]
D4
F2
D3
D2
D1
D6
D5
F4
F3
H5
J5
74LS273
74LS174
74LS283
flip flops
hex flip flop
SCR1X0CS
SCR1X1CS
SCR2X0CS
SCR2X1CS
VCC
VSS
XHMSB
74LS04
74LS08
74LS86
74LS10
74LS00
SCRH_CLK
SCR1S0
SCR_CLK
MAP2SCR1
MAP2SCR0
HS_X3
HS_X4
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JOTEGO
Sheet: /video/h_scroll/
File: h_scroll1.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 3/28



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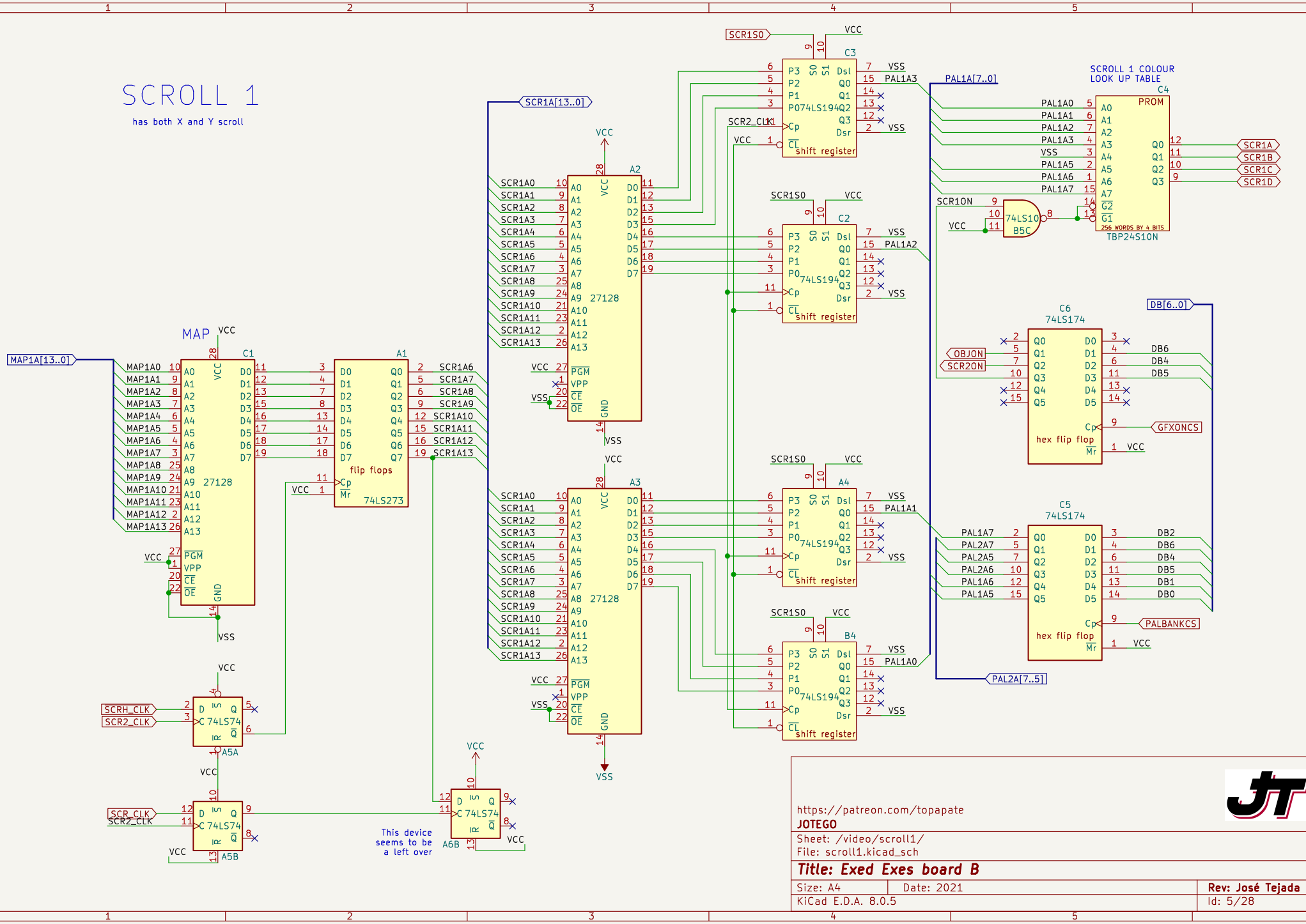
JOTEGO

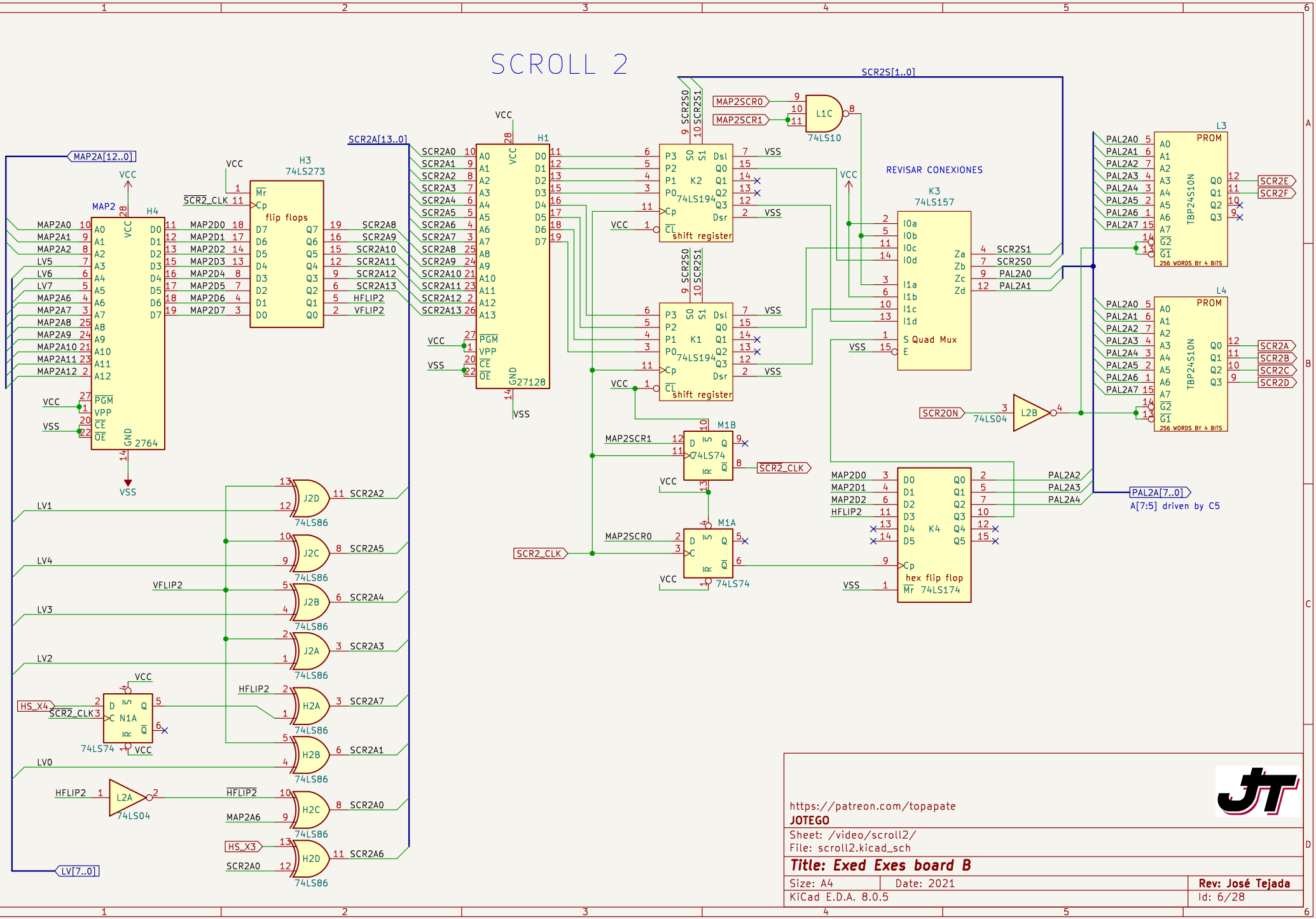
Sheet: /video/v_scroll/

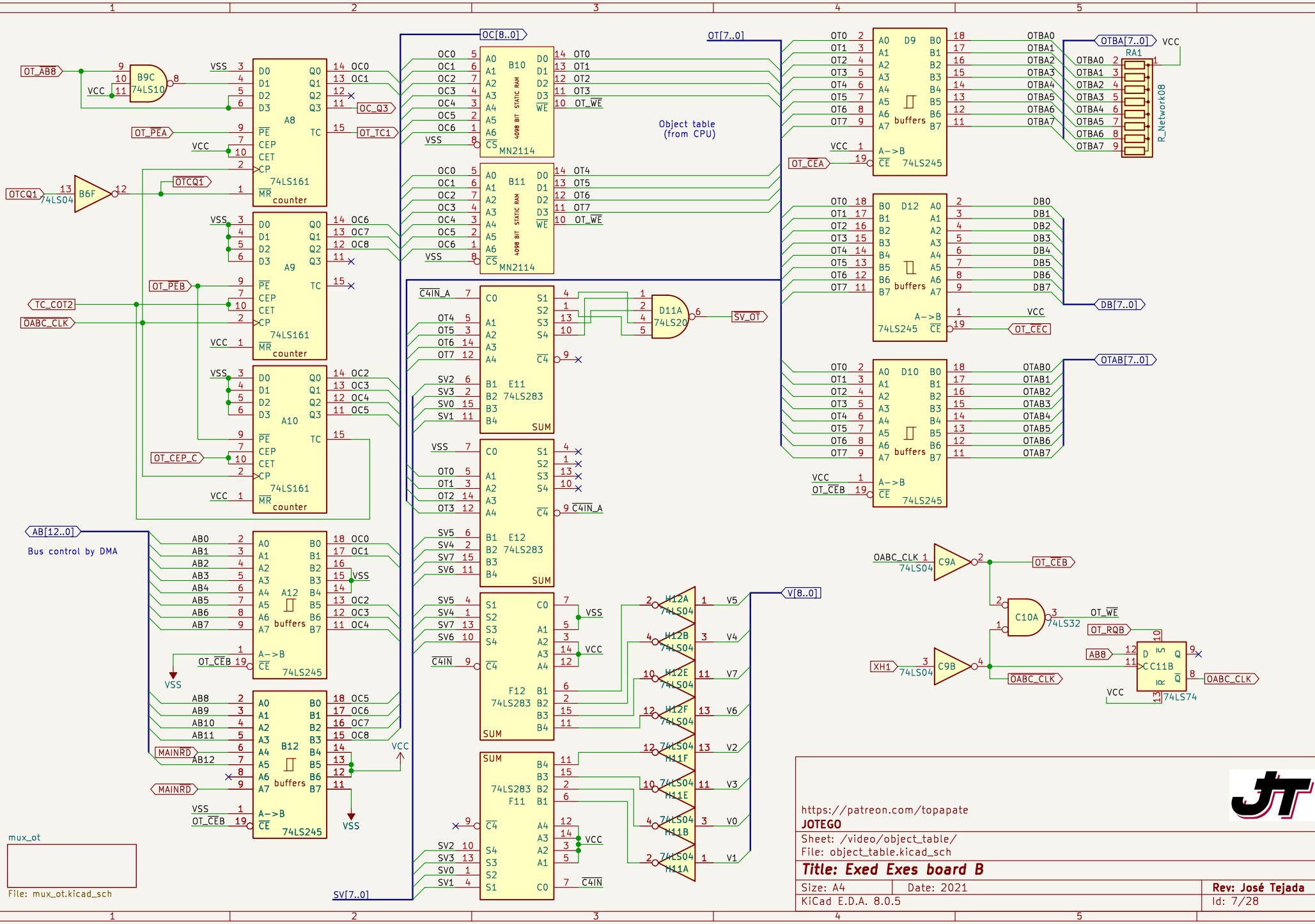
File: v_scroll.kicad_sch

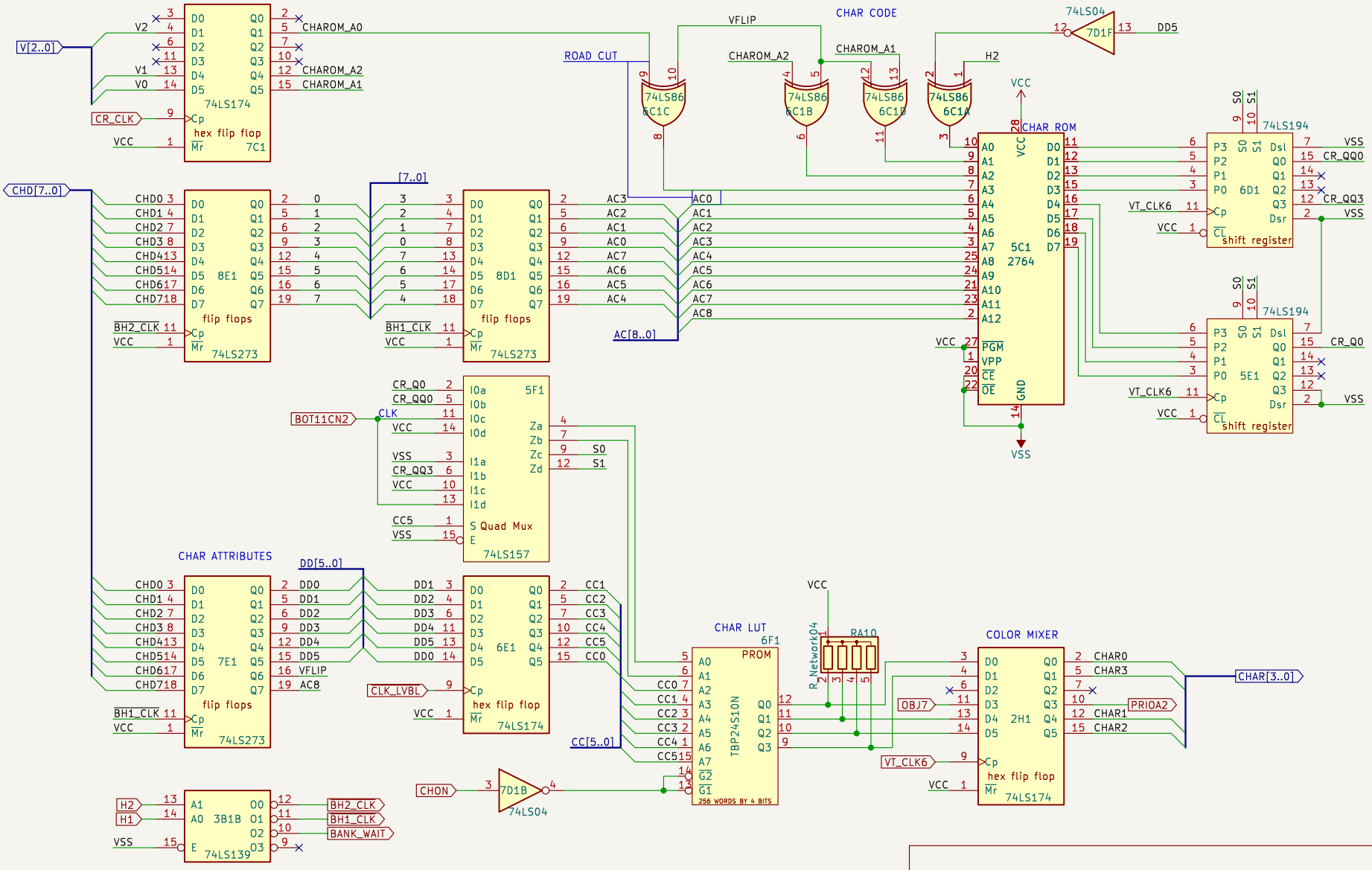
Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 4/28


SCROLL 1
has both X and Y scroll
MAP
MAP1A[13..0]
C1
27128
A1
74LS273
flip flops
SCR1A[13..0]
A2
A3
C3
C2
A4
B4
74LS194
shift register
SCR1S0
SCR2_CLK
PAL1A[7..0]
PAL2A[7..5]
SCROLL 1 COLOUR
LOOK UP TABLE
C4
PROM
TBP24S10N
256 WORDS BY 4 BITS
SCR1ON
74LS10
B5C
C6
74LS174
hex flip flop
C5
OBJON
SCR2ON
DB[6..0]
GFXONCS
PALBANKCS
SCRH_CLK
SCR2_CLK
SCR_CLK
SCR2_CLK
74LS74
A5A
A5B
A6B
This device
seems to be
a left over
https://patreon.com/topapate
JOTEGO
Sheet: /video/scroll1/
File: scroll1.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 5/28


SCROLL 2
SCR2S[1..0]
MAP2A[12..0]
SCR2A[13..0]
REVISAR CONEXIONES
MAP2
H4
2764
H3
74LS273
flip flops
H1
27128
K3
74LS157
S Quad Mux
L3
L4
PROM
TBP24S10N
256 WORDS BY 4 BITS
K4
hex flip flop
74LS174
74LS194
shift register
74LS74
74LS86
74LS04
74LS10
M1A
M1B
L1C
L2A
L2B
J2D
J2C
J2B
J2A
H2A
H2B
H2C
H2D
SCR2_CLK
HS_X4
HS_X3
LV[7..0]
LV0
LV1
LV2
LV3
LV4
VFLIP2
HFLIP2
MAP2SCR0
MAP2SCR1
SCR2ON
PAL2A[7..0]
A[7:5] driven by C5
https://patreon.com/topapate
JOTEGO
Sheet: /video/scroll2/
File: scroll2.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 6/28


OT_AB8
B9C
74LS10
VCC
VSS
A8
74LS161
counter
OT_PEA
OTCQ1
B6F
74LS04
A9
OT_PEB
TC_COT2
OABC_CLK
A10
OT_CEP_C
AB[12..0]
Bus control by DMA
A12
buffers
A->B
74LS245
OT_CEB
MAINRD
B12
mux_ot
File: mux_ot.kicad_sch
OC[8..0]
B10
B11
4098 BIT STATIC RAM
MN2114
OT[7..0]
Object table
(from CPU)
OT_WE
C4IN_A
D11A
74LS20
SV_OT
E11
74LS283
SUM
E12
F12
C4IN
SV[7..0]
V[8..0]
H12A
H12B
H12E
H12F
H11F
H11E
H11B
H11A
D9
D12
D10
OTBA[7..0]
OTAB[7..0]
DB[7..0]
RA1
R_Network08
C9A
C9B
C10A
74LS32
XH1
OT_RQB
C11B
74LS74
https://patreon.com/topapate
JOTEGO
Sheet: /video/object_table/
File: object_table.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 7/28



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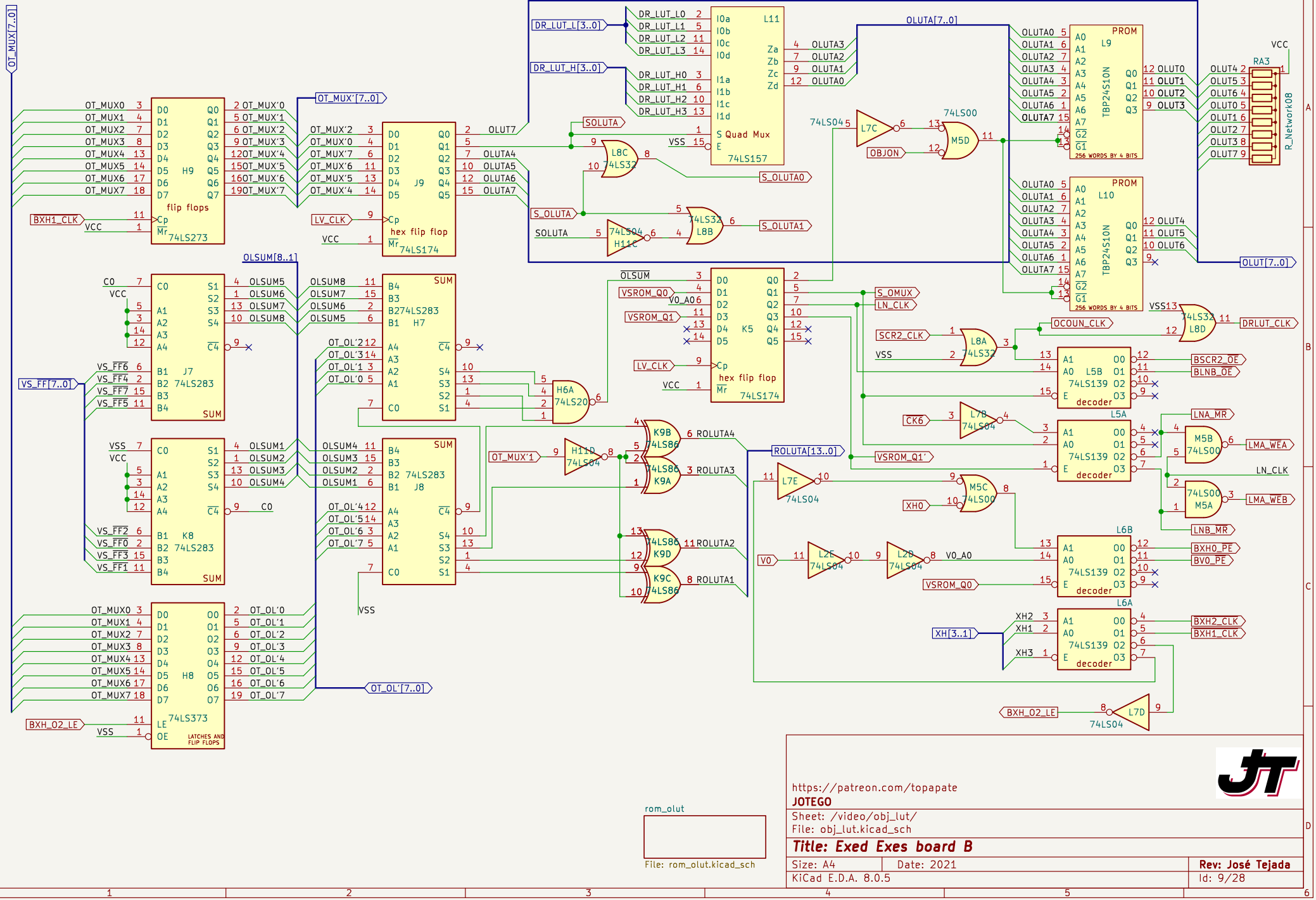
JOTEGO

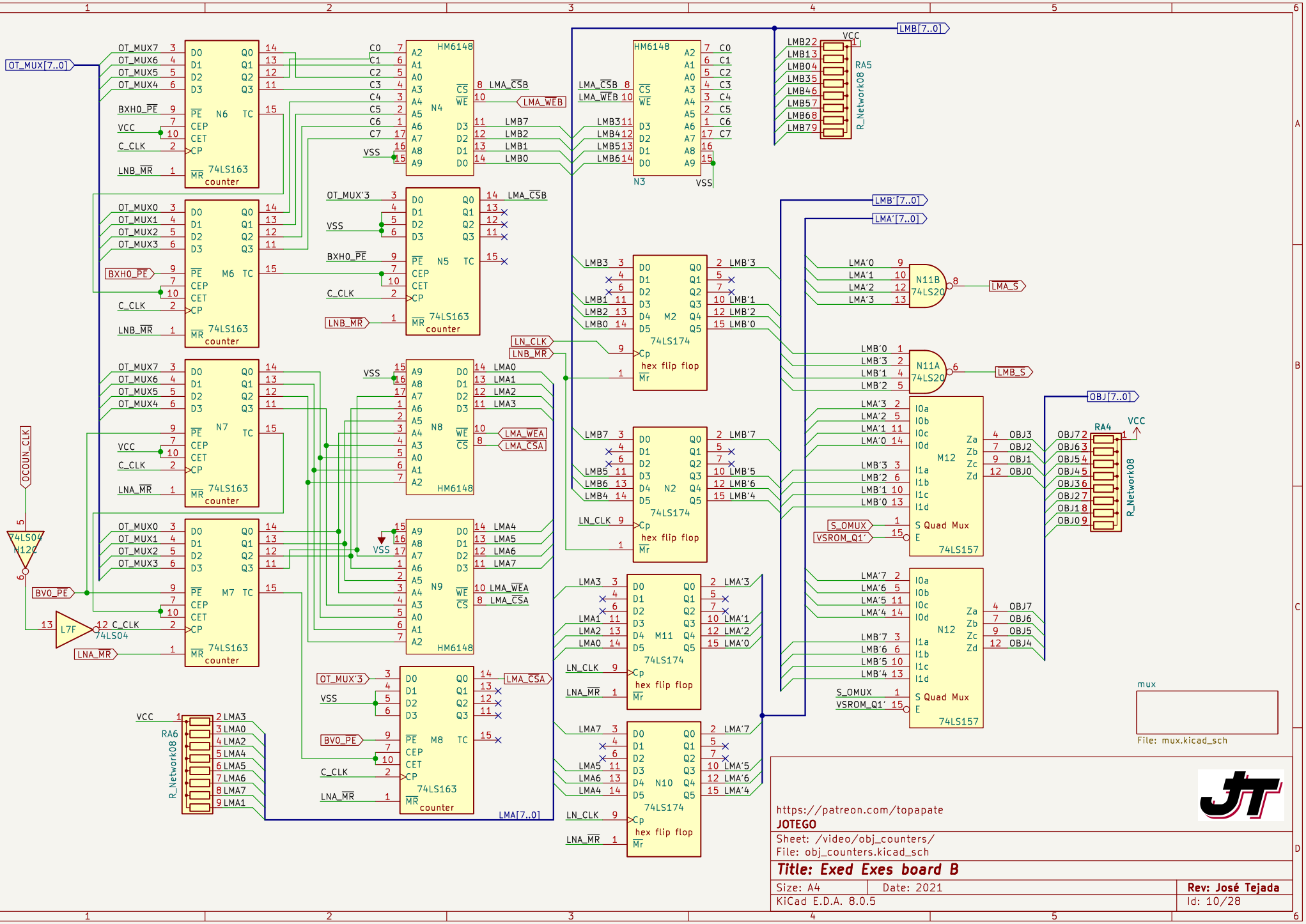
Sheet: /video/char/

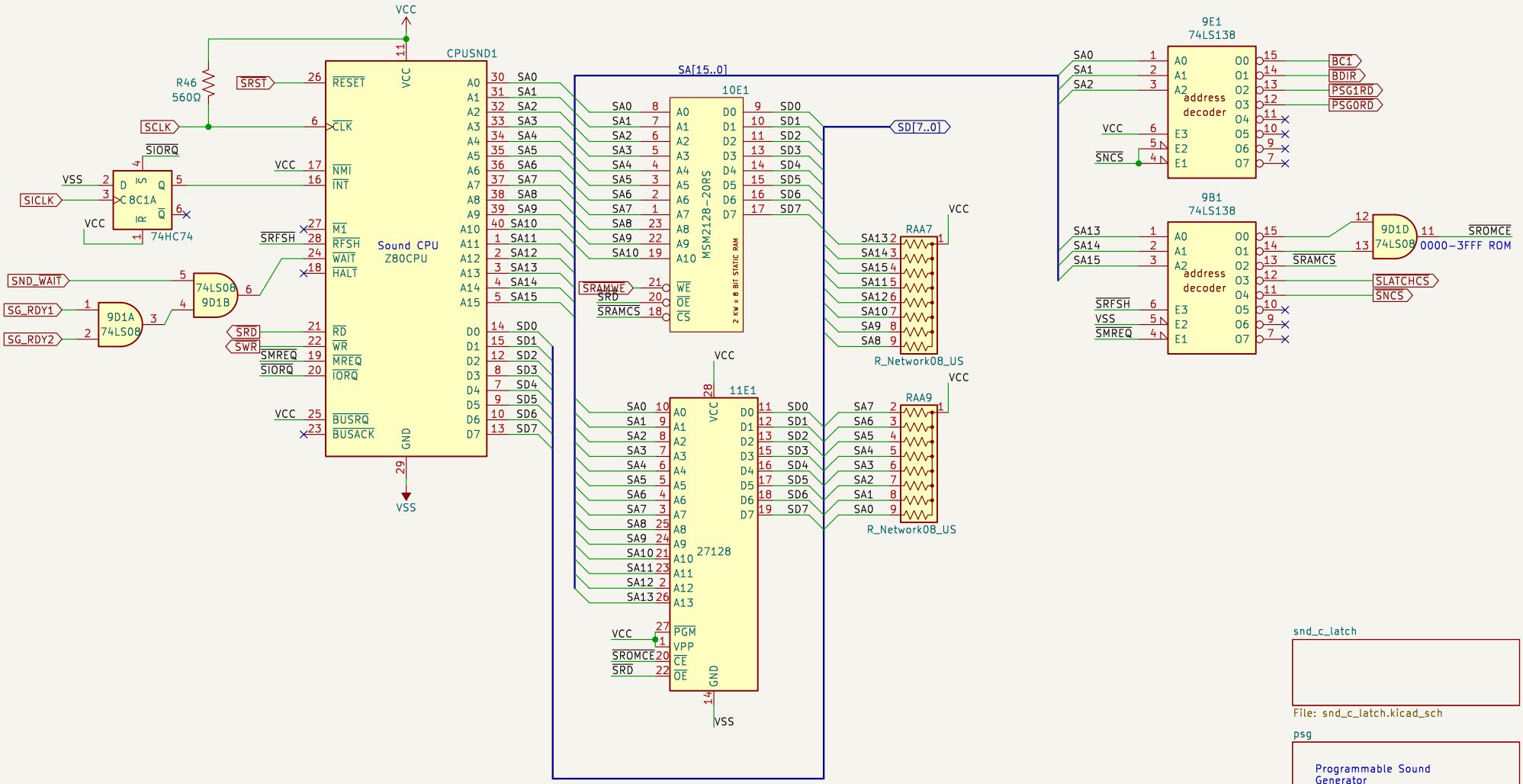
File: char.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 8/28


OT_MUX[7..0]
DR_LUT_L[3..0]
DR_LUT_H[3..0]
OLUTA[7..0]
OLUT[7..0]
OT_MUX'[7..0]
OLSUM[8..1]
VS_FF[7..0]
OT_OL'[7..0]
ROLUTA[13..0]
XH[3..1]
74LS373
74LS273
flip flops
hex flip flop
74LS174
74LS157
Quad Mux
74LS283
74LS139
decoder
TBP24S10N
PROM
256 WORDS BY 4 BITS
R_Network08
LATCHES AND FLIP FLOPS
https://patreon.com/topapate
JOTEGO
Sheet: /video/obj_lut/
File: obj_lut.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 9/28
rom_olut
File: rom_olut.kicad_sch


OT_MUX[7..0]
LMB[7..0]
LMB'[7..0]
LMA'[7..0]
OBJ[7..0]
LMA[7..0]
OCOUN_CLK
LN_CLK
LNB_MR
LNA_MR
BXH0_PE
BV0_PE
C_CLK
VCC
VSS
LMA_CSB
LMA_WEB
LMA_WEA
LMA_CSA
LMA_S
LMB_S
S_OMUX
VSROM_Q1'
N4
N5
N6
N7
N8
N9
N10
N11
N12
M2
M6
M7
M8
M11
M12
74LS163
counter
74LS174
hex flip flop
HM6148
74LS157
Quad Mux
74LS20
74LS04
R_Network08
RA4
RA5
RA6
mux
File: mux.kicad_sch
https://patreon.com/topapate
JOTEGO
Sheet: /video/obj_counters/
File: obj_counters.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 10/28



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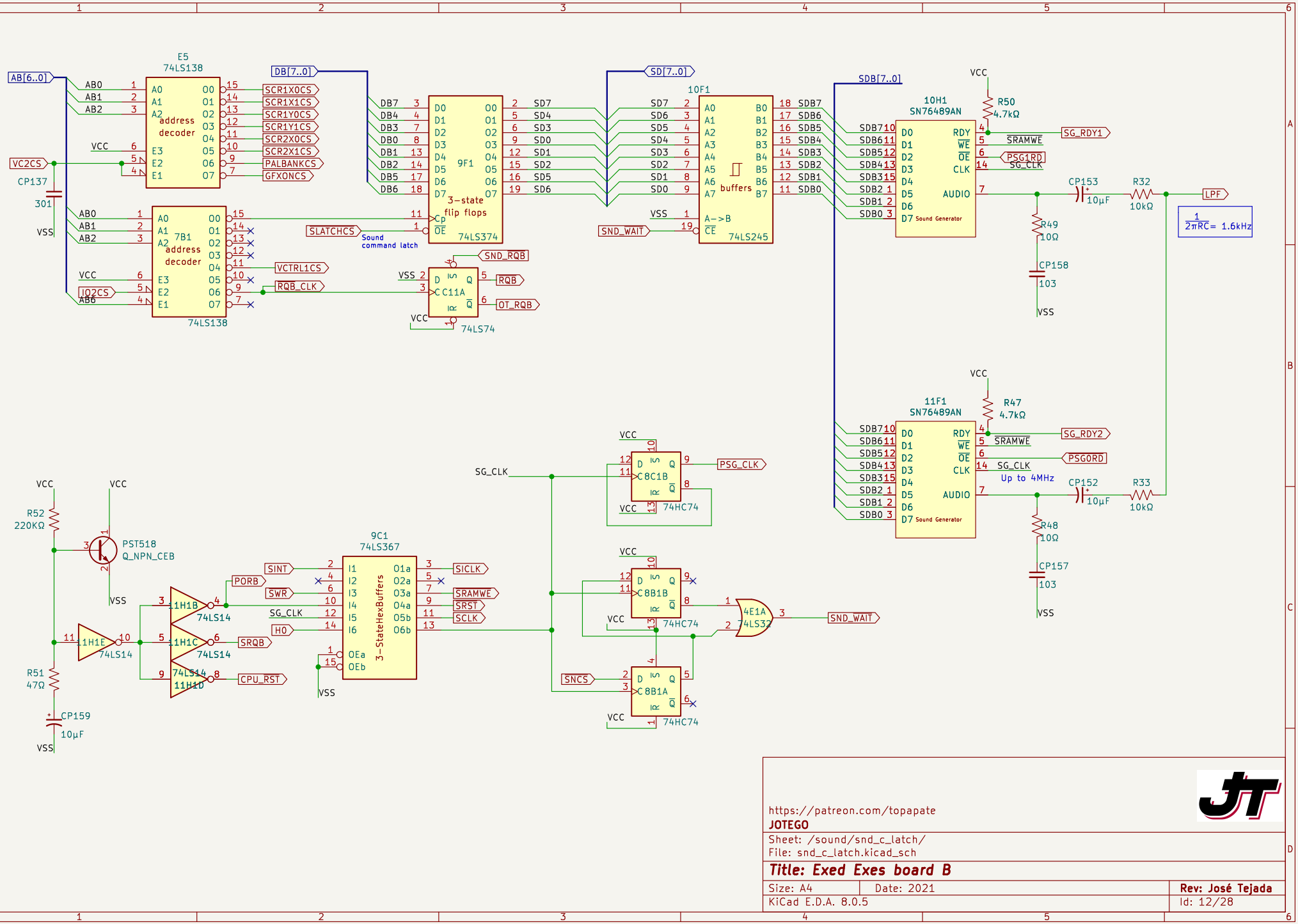
JOTEGO

Sheet: /sound/

File: sound.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 11/28



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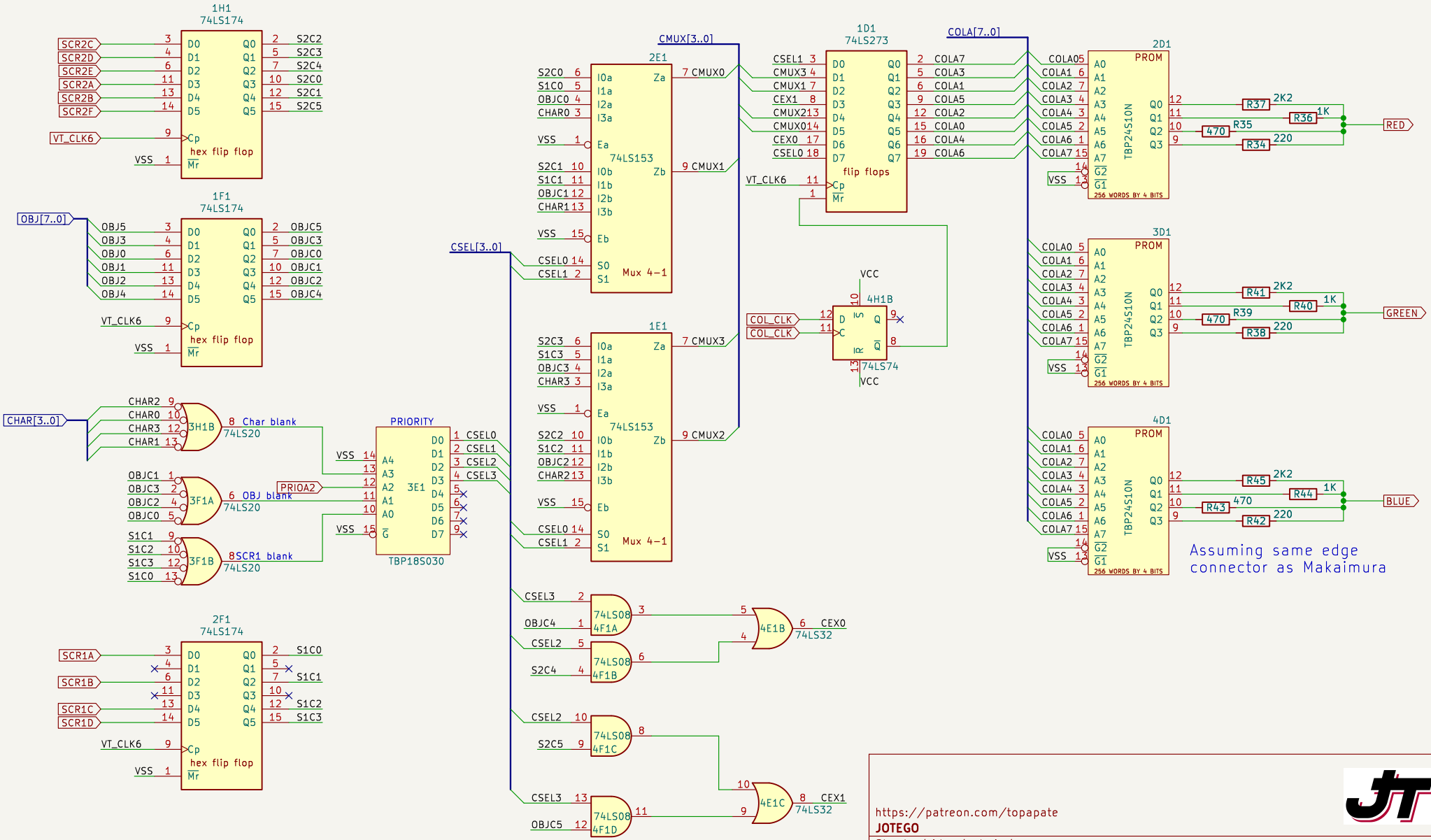
JOTEGO

Sheet: /sound/snd_c_latch/
File: snd_c_latch.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 12/28

COLOR PROMS



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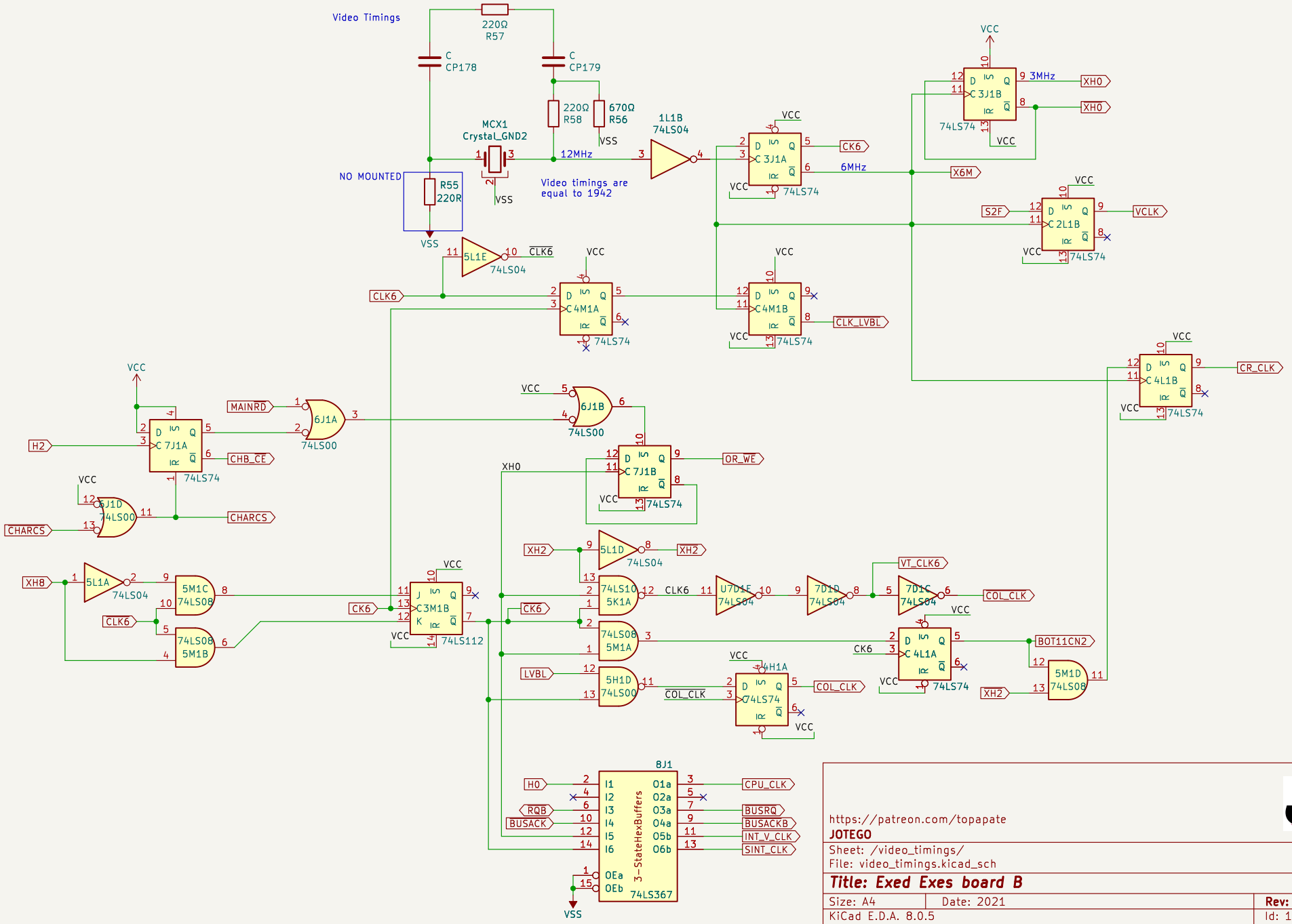
JOTEGO

Sheet: /video/colmix/

File: colmix.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 13/28



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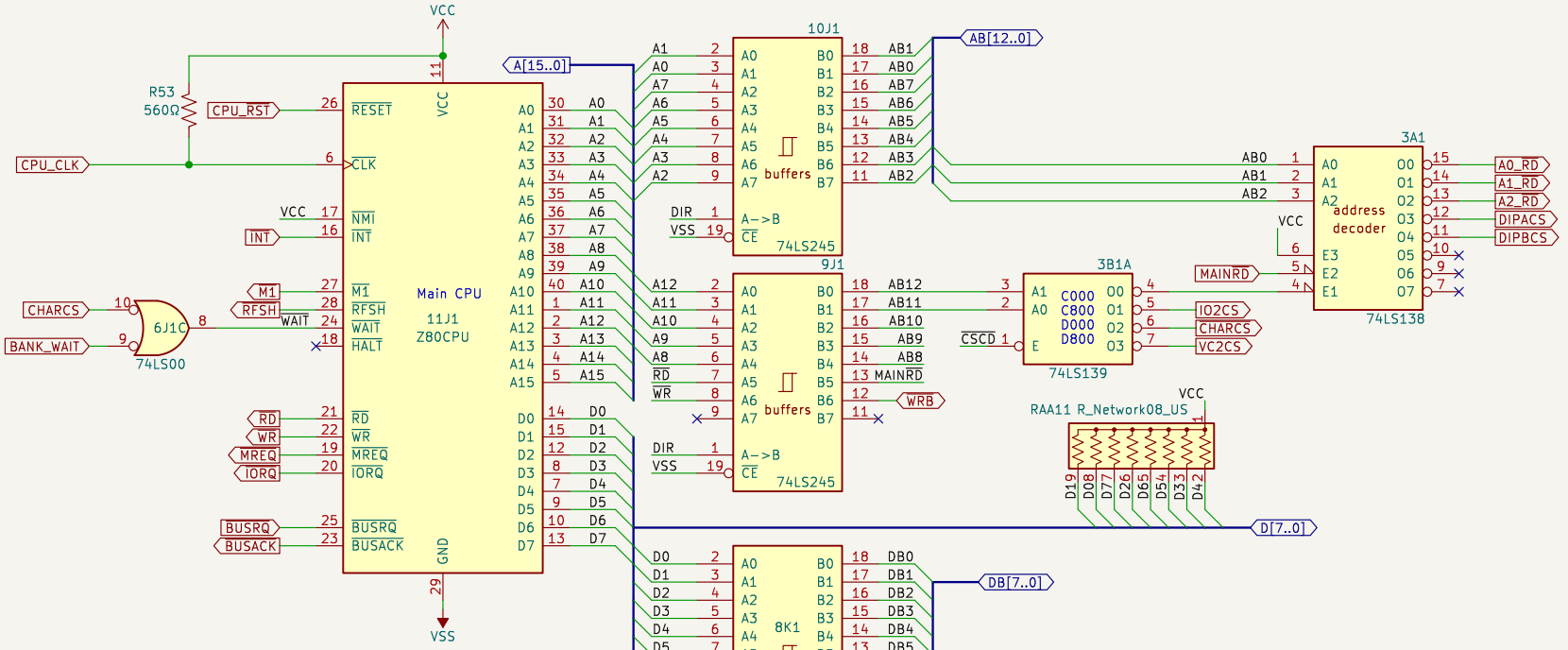
JOTEGO

Sheet: /video_timings/

File: video_timings.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 14/28



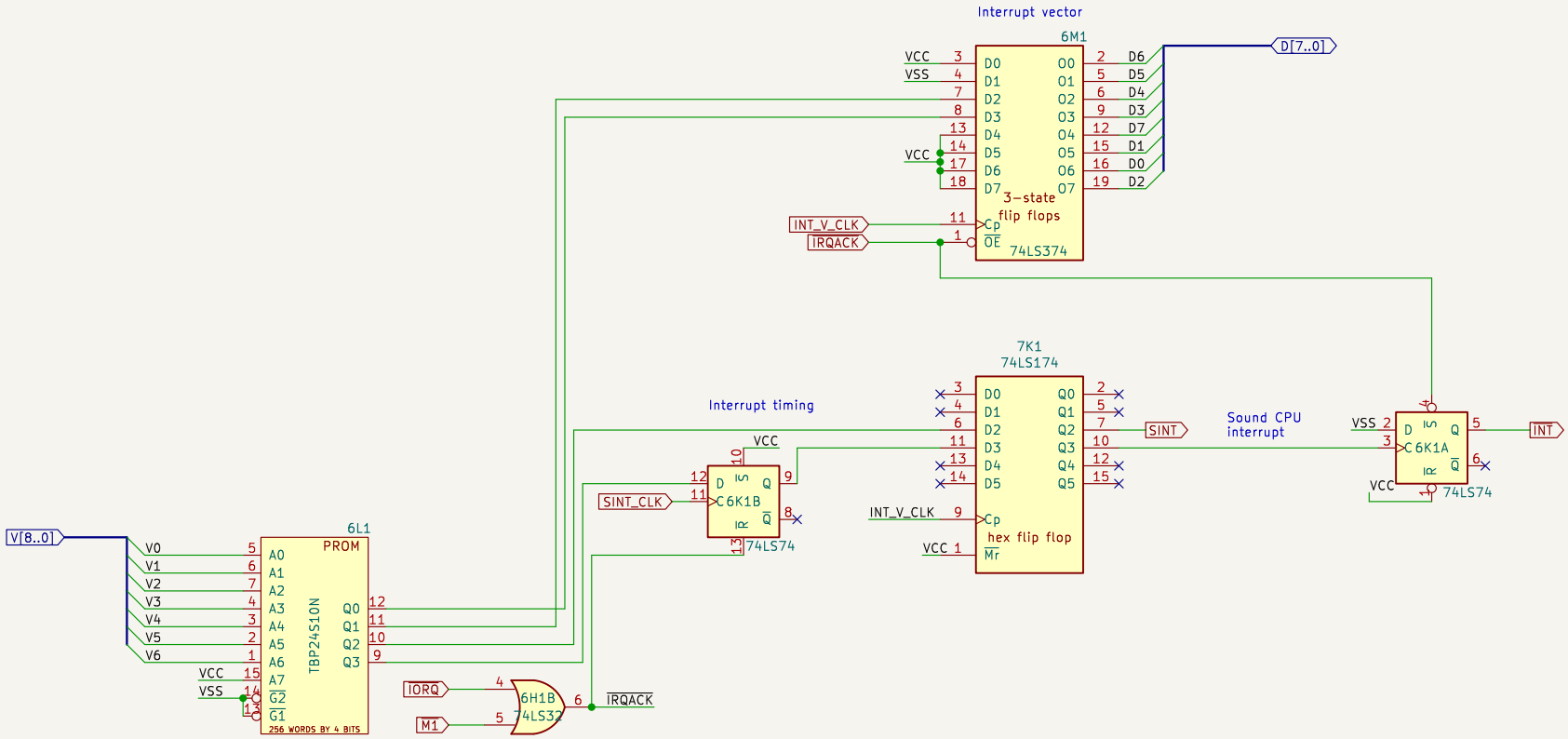
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JOTEGO

Sheet: /main/
File: main.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 15/28



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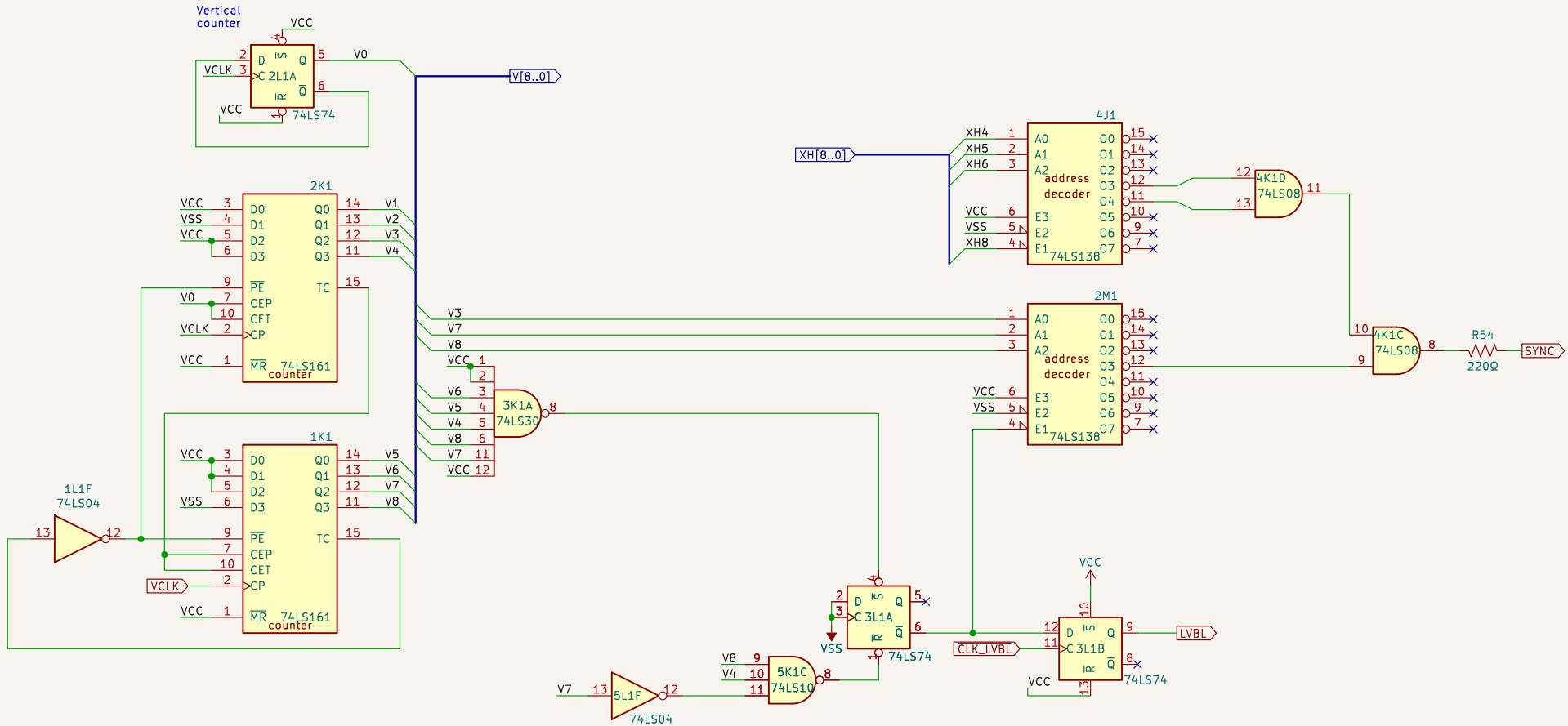
JOTEGO

Sheet: /interruption/

File: interruption.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 16/28



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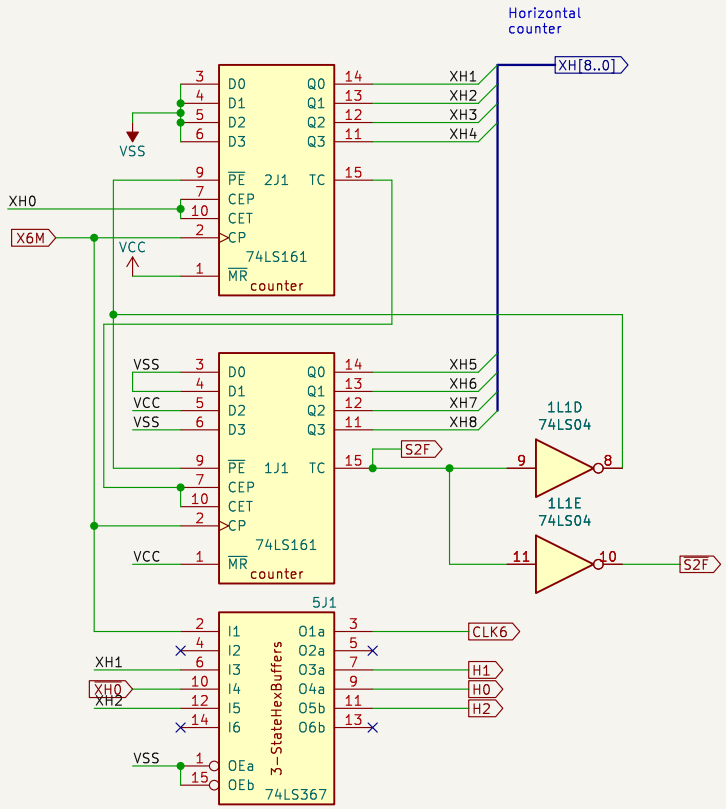
JOTEGO

Sheet: /v_counter/

File: v_counter.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 17/28



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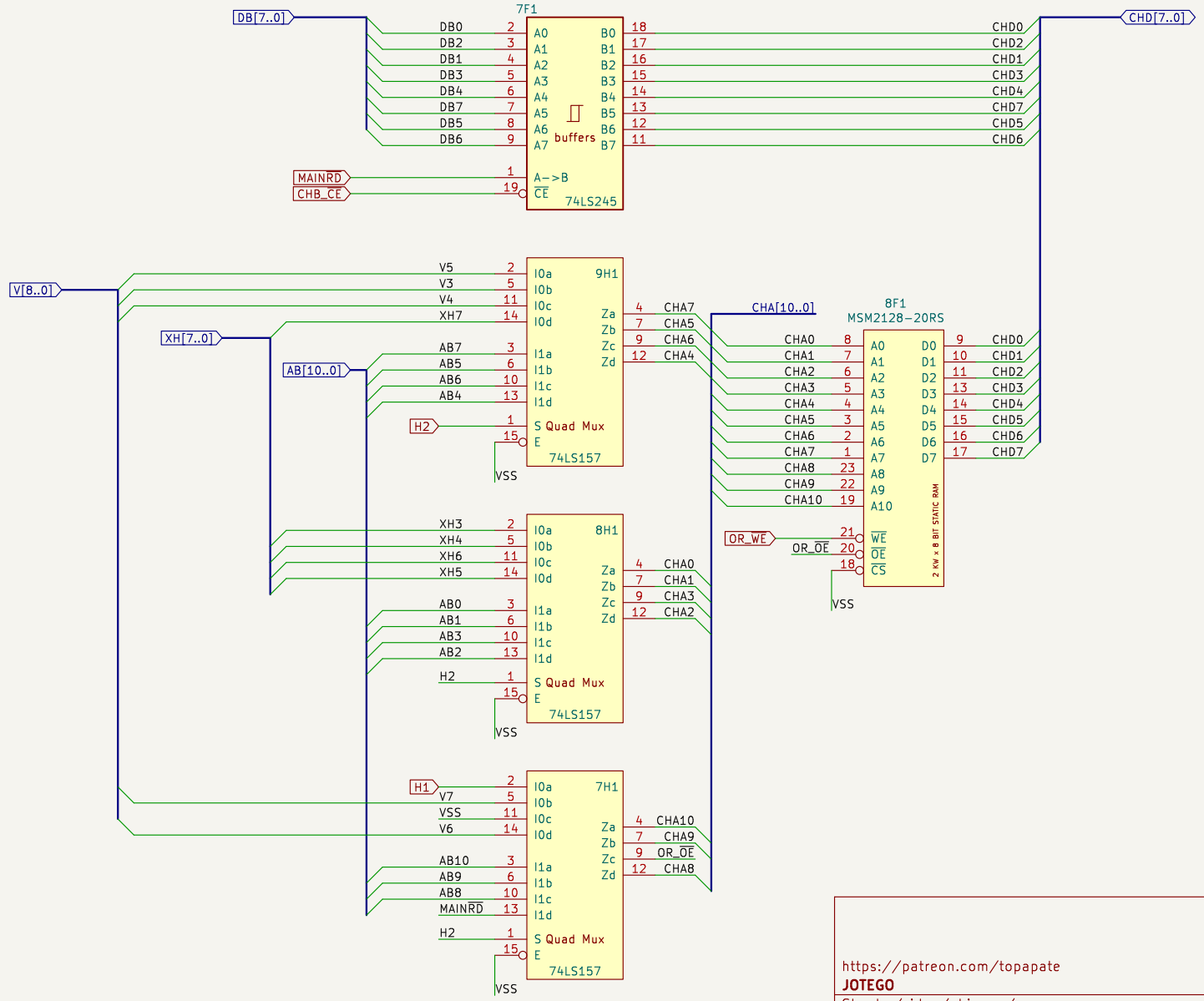
JOTEGO

Sheet: /h_counter/

File: h_counter.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 18/28



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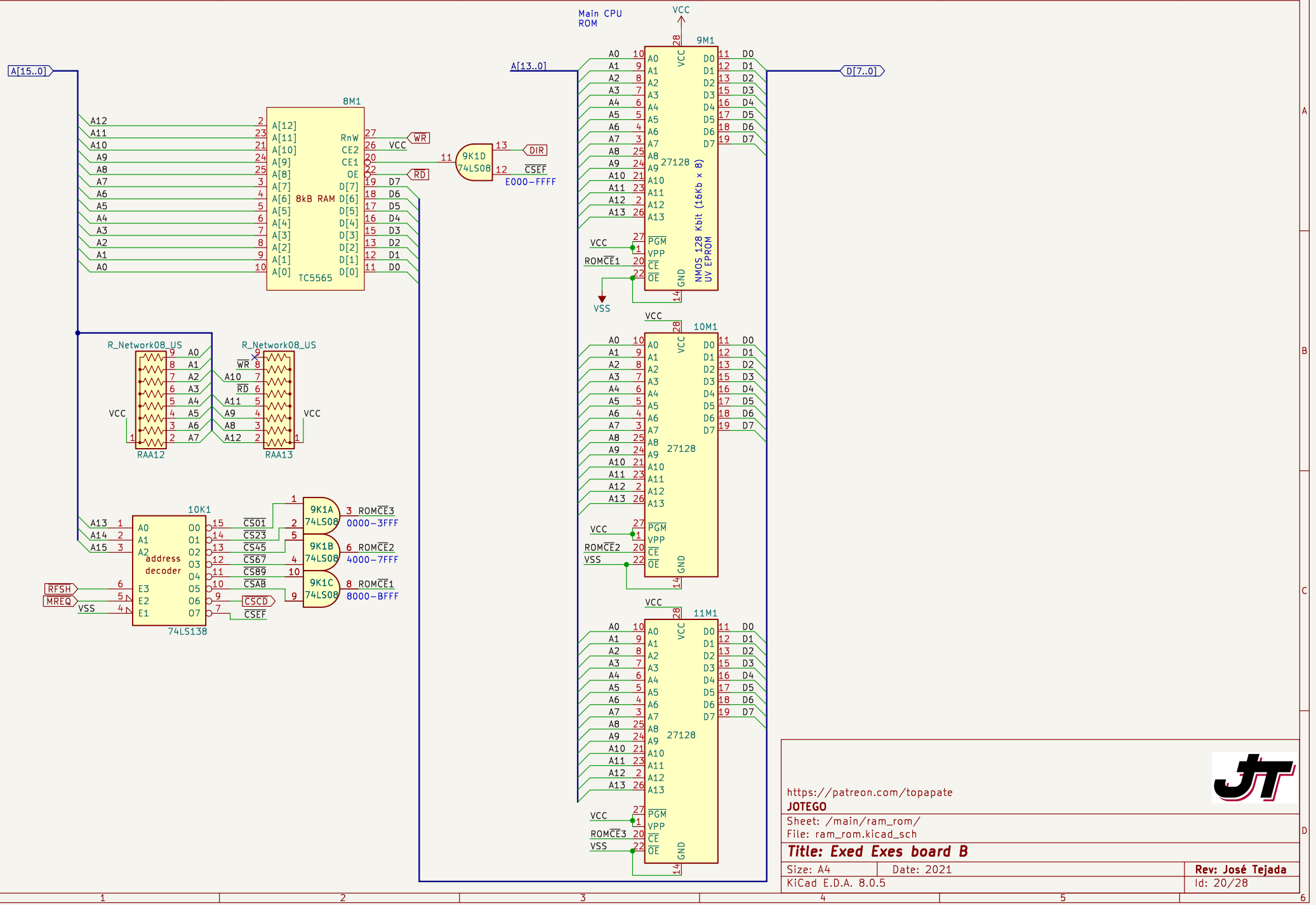
JOTEGO

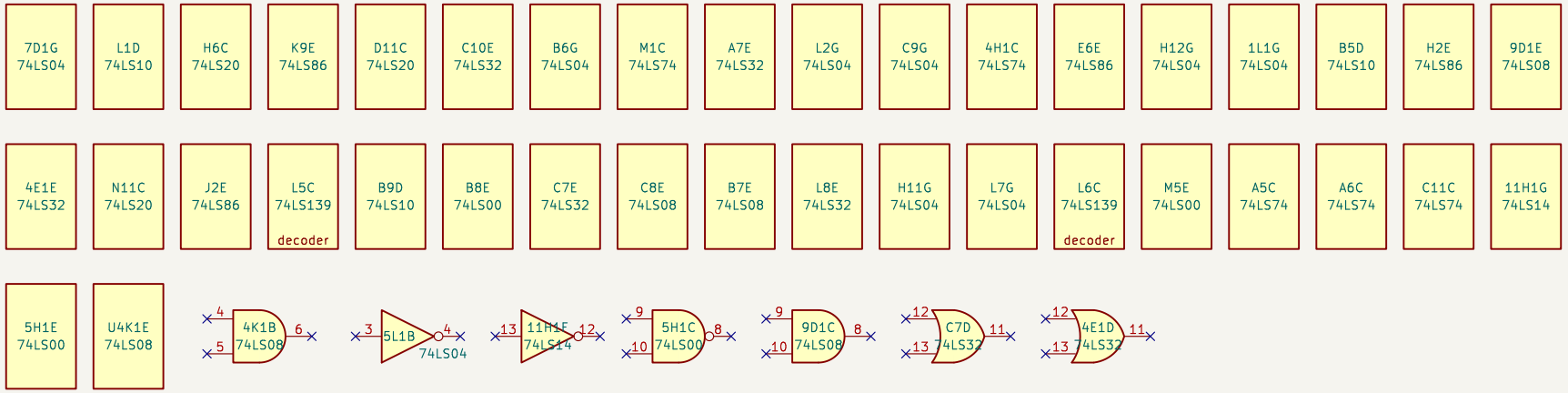
Sheet: /video/obj_ram/

File: obj_ram.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 19/28


Main CPU
ROM
A[15..0]
A[13..0]
D[7..0]
8M1
8kB RAM
TC5565
WR
RD
DIR
CSEF
E000-FFFF
9K1D
74LS08
9M1
10M1
11M1
27128
NMOS 128 Kbit (16Kb x 8)
UV EPROM
ROMCE1
ROMCE2
ROMCE3
R_Network08_US
RAA12
RAA13
10K1
74LS138
address
decoder
RFSH
MREQ
9K1A
9K1B
9K1C
ROMCE3 0000-3FFF
ROMCE2 4000-7FFF
ROMCE1 8000-BFFF
https://patreon.com/topapate
JOTEGO
Sheet: /main/ram_rom/
File: ram_rom.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.5
Id: 20/28



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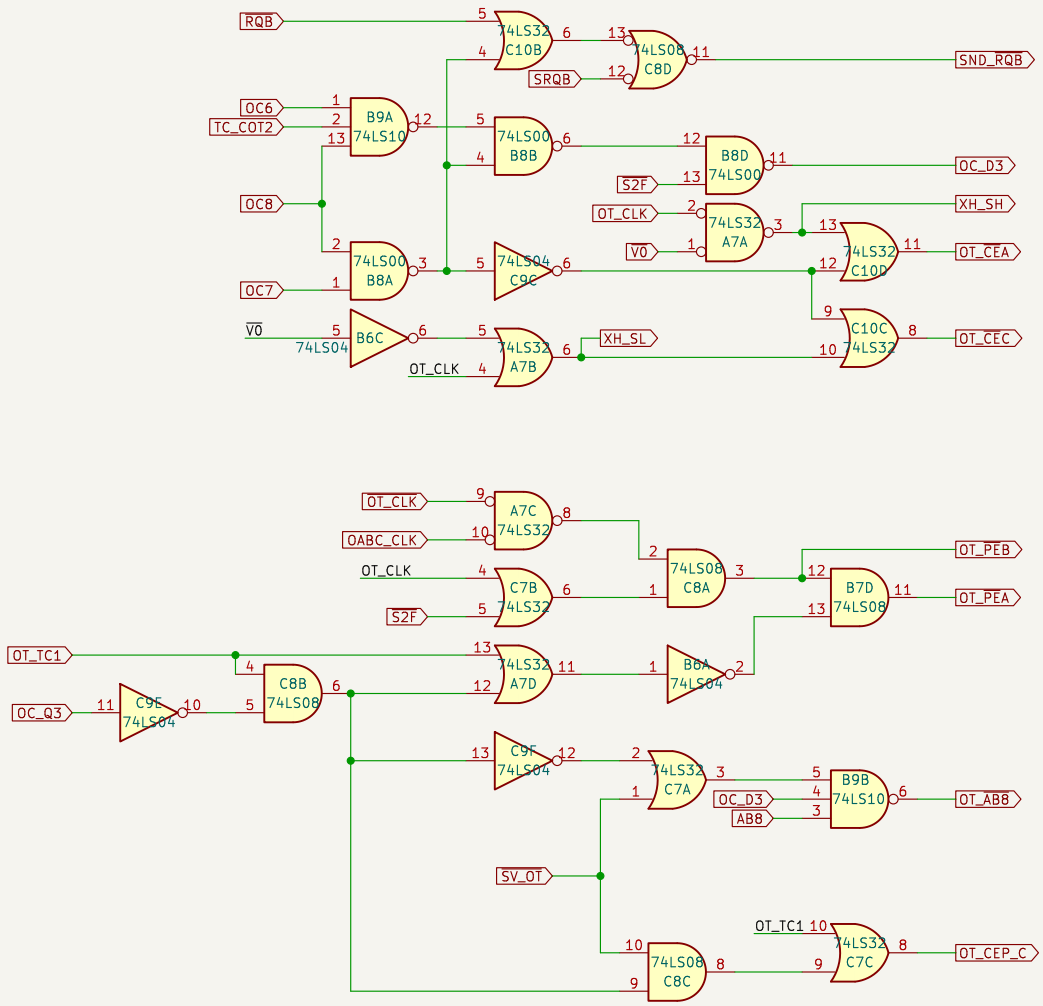
JOTEGO

Sheet: /last_u/

File: last_u.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 20/28



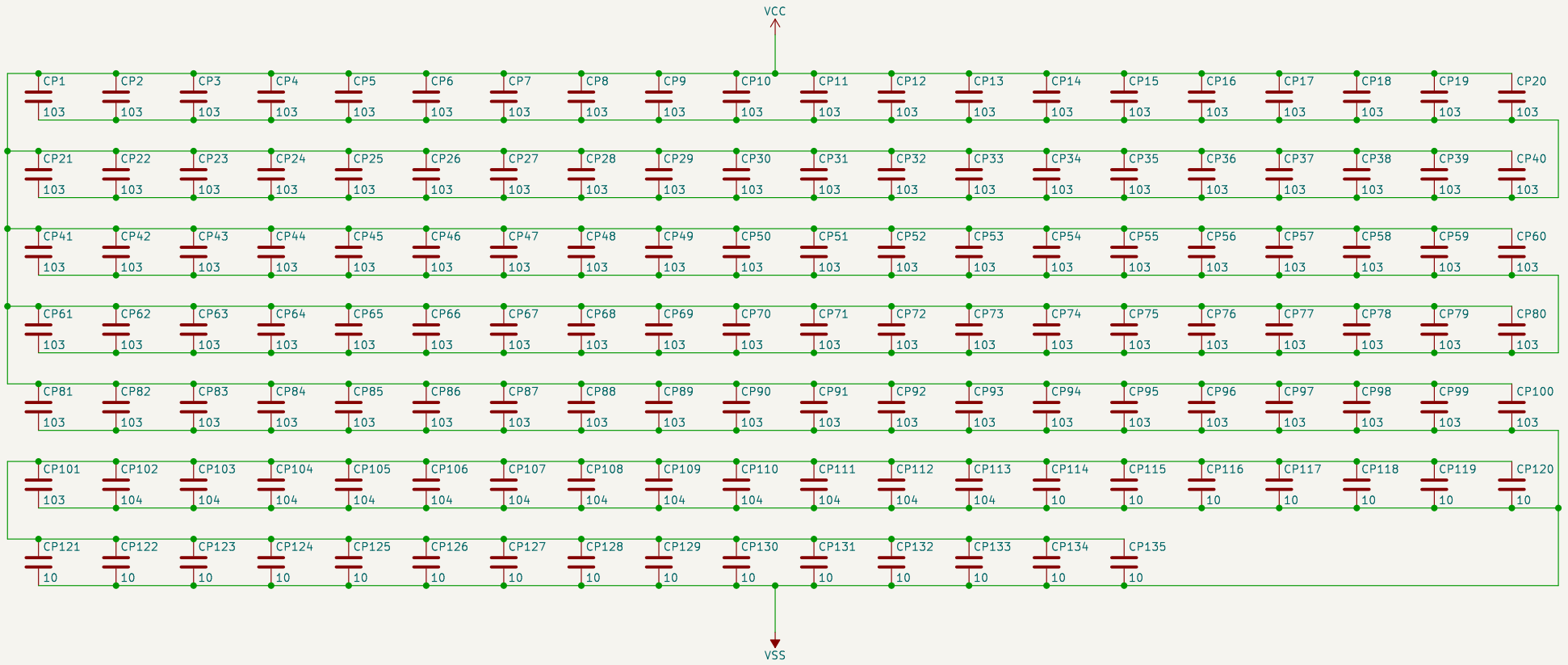
https://patreon.com/topapate

JOTEGO

Sheet: /video/logics_gates/
File: logics_gates.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 21/28



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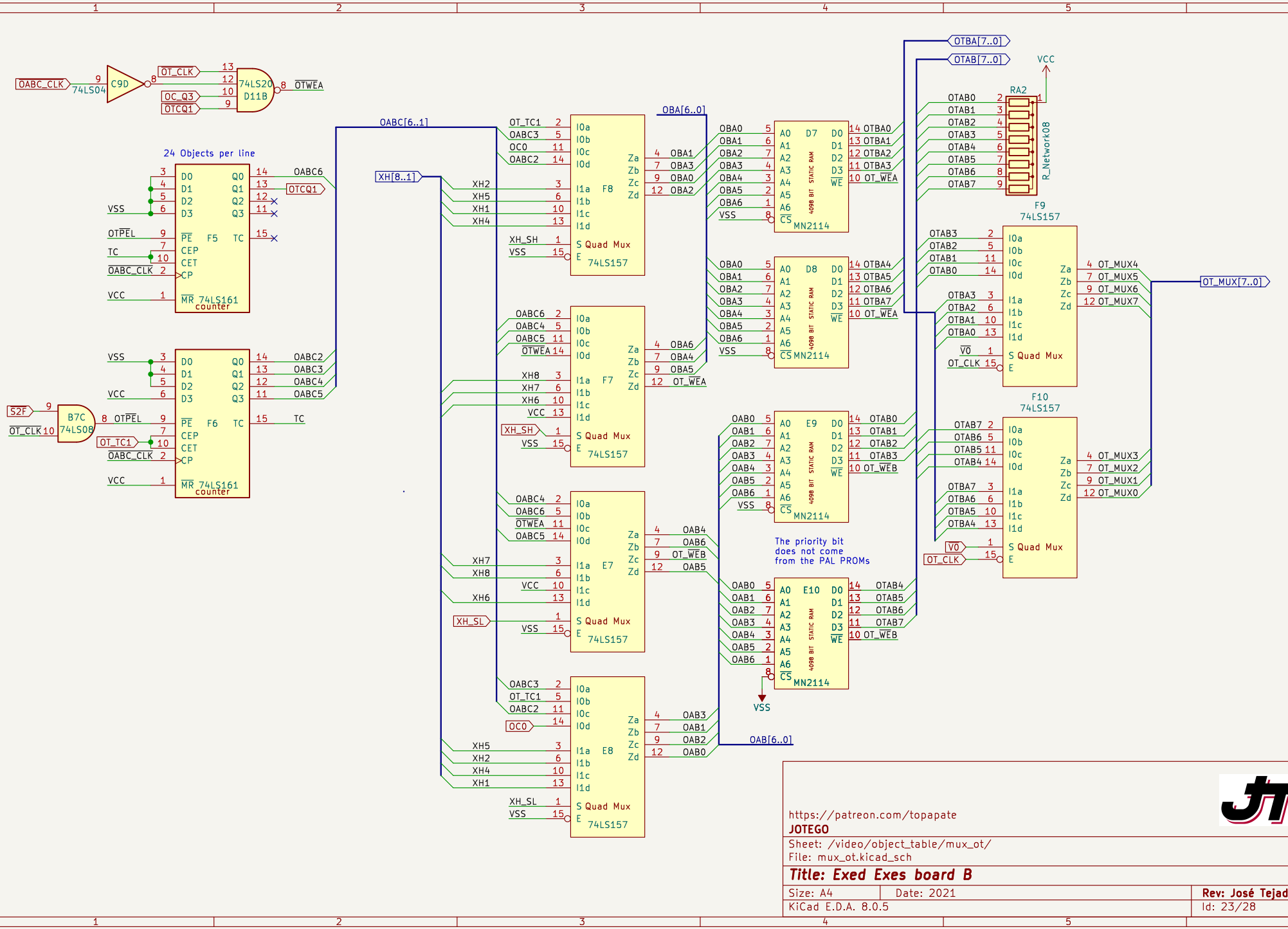
JOTEGO

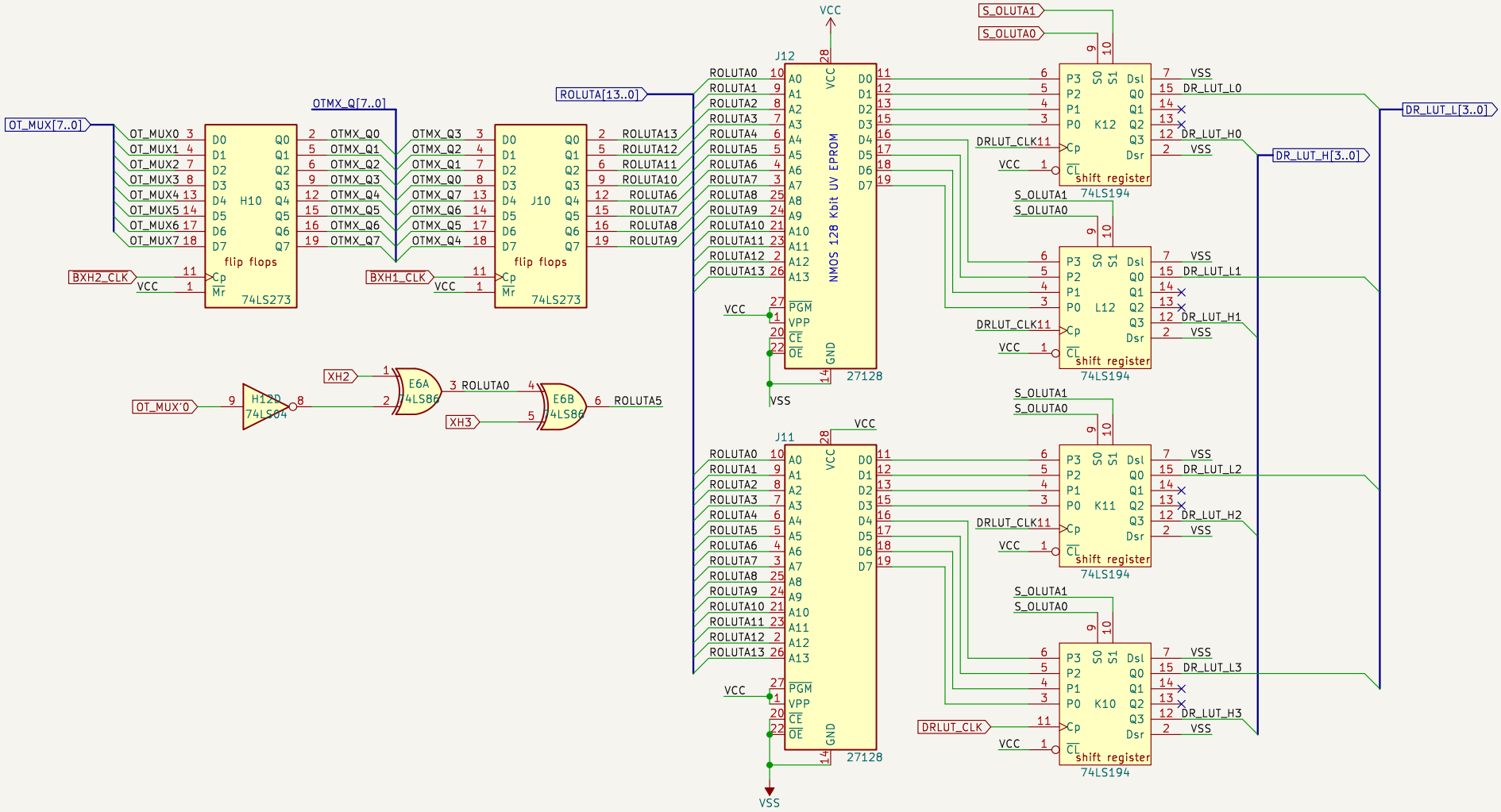
Sheet: /capacitors/

File: capacitors.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 22/28


OABC_CLK
74LS04
C9D
OT_CLK
OC_Q3
OTCQ1
74LS20
D11B
OTWEA
24 Objects per line
OABC[6..1]
XH[8..1]
OABC6
OTCQ1
OABC2
OABC3
OABC4
OABC5
TC
S2F
OT_CLK
B7C
74LS08
OTPEL
OT_TC1
74LS161
counter
F5
F6
F7
F8
F9
F10
74LS157
Quad Mux
OBA[6..0]
OAB[6..0]
OTBA[7..0]
OTAB[7..0]
VCC
RA2
R_Network08
MN2114
4098 BIT STATIC RAM
E7
E8
E9
E10
The priority bit
does not come
from the PAL PROMs
OT_MUX[7..0]
OT_MUX7
OT_MUX6
OT_MUX5
OT_MUX4
OT_MUX3
OT_MUX2
OT_MUX1
OT_MUX0
OT_WEA
OT_WEB
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JOTEGO
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Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.5
Rev: José Tejada
Id: 23/28



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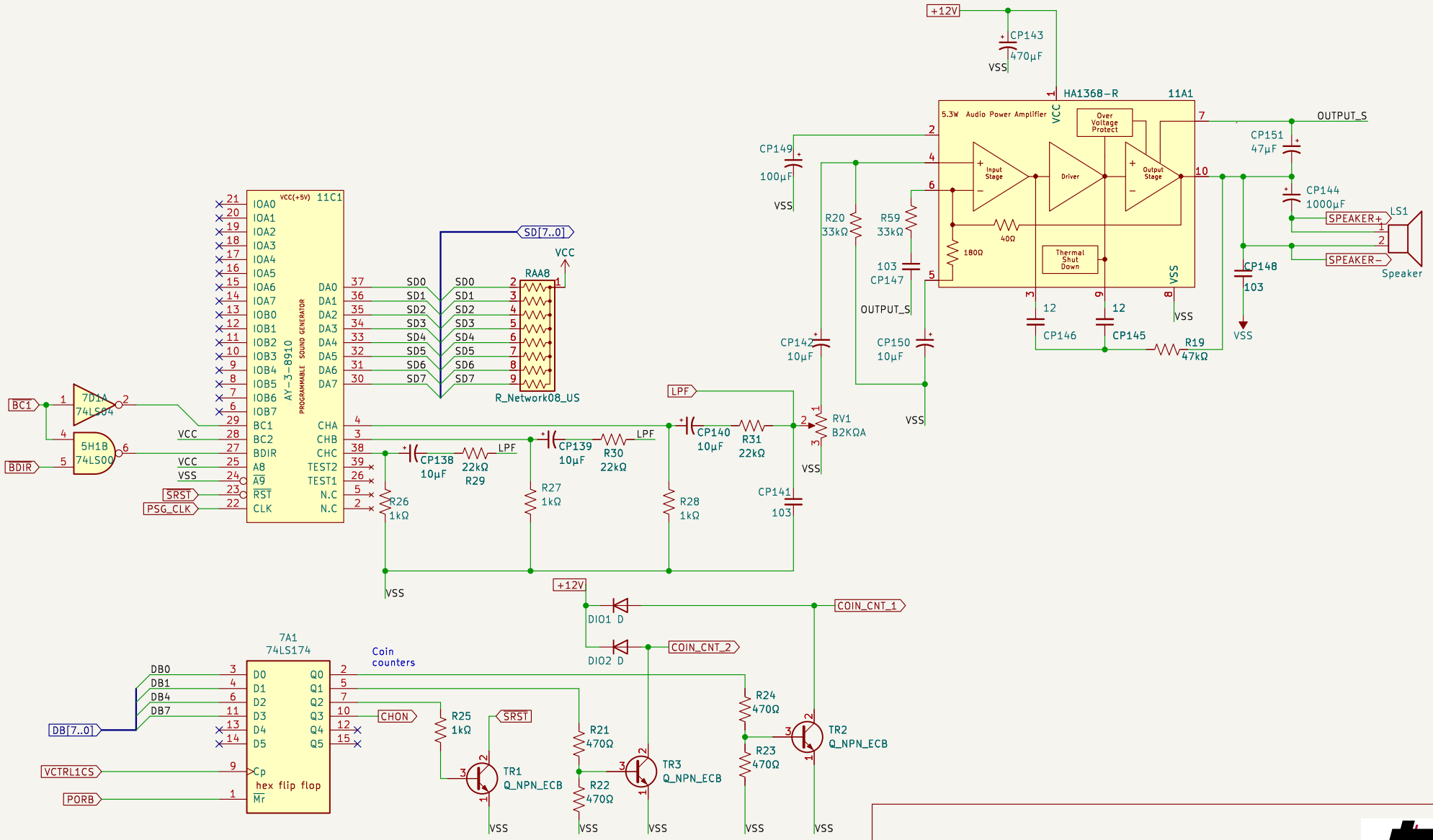
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
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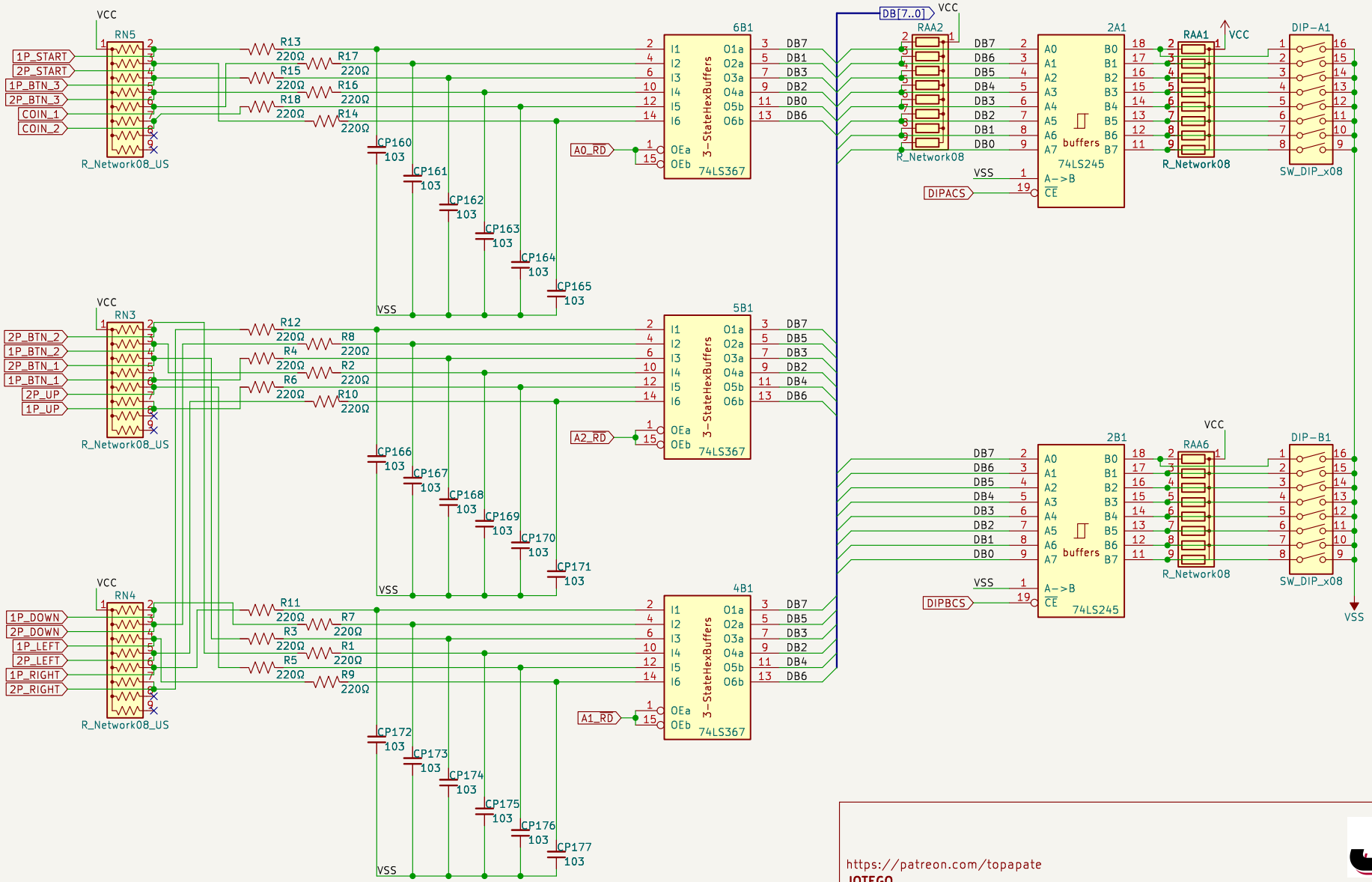
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 26/28



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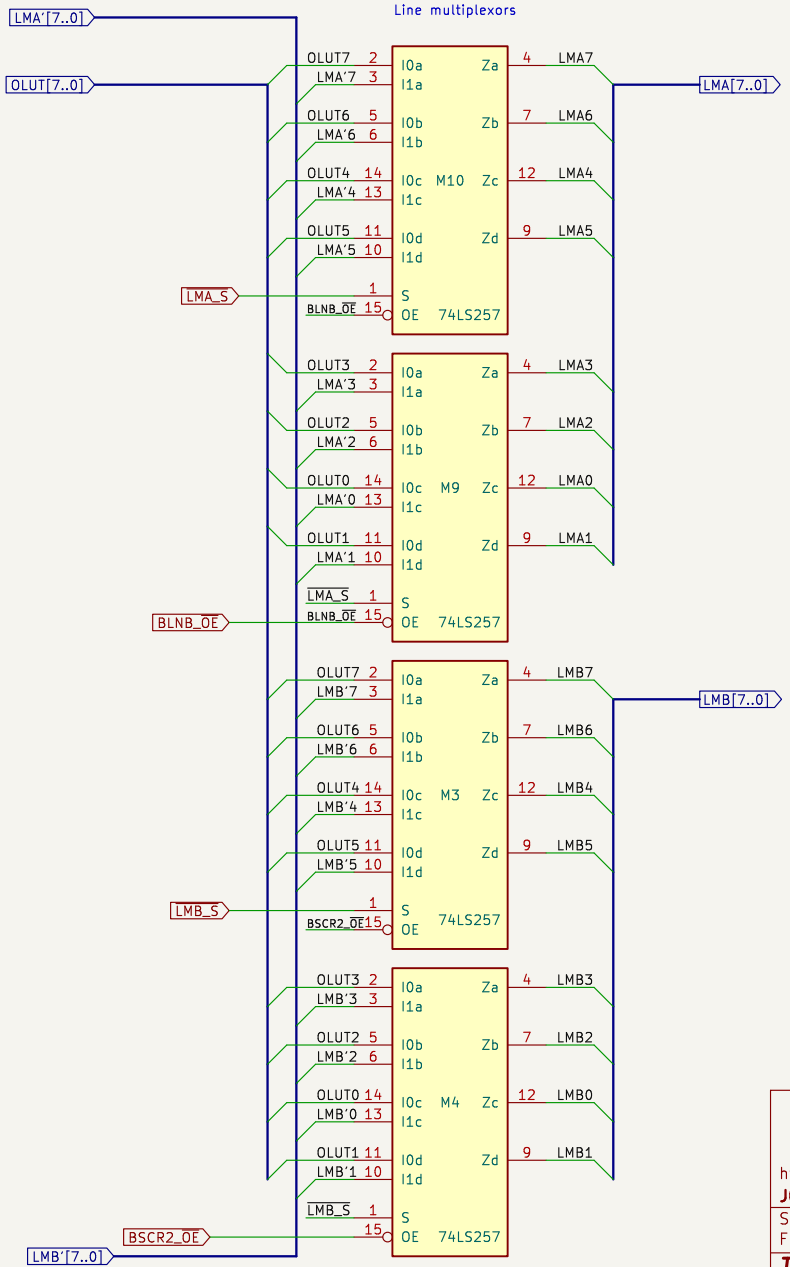
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 26/28



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JOTEGO

Sheet: /video/obj_counters/mux/
File: mux.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.5		Id: 28/28